When Do Single-Pass Action Heads Help? A Factorized Decoder for Vision–Language–Action Models

An operator-inspired branch–trunk head, evaluated across four backbone scales (88 M–3.3 B) on LIBERO and LIBERO-Plus

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Abstract

Vision–language–action (VLA) models decode continuous robot actions with heavy generative heads—flow-matching or diffusion decoders that run an iterative sampling loop at every control step. We ask a simple question: can a lightweight, single-pass head replace them? We introduce a **factorized single-pass action head** — a branch–trunk decoder inspired by operator learning, though we make no theoretical claim from it — that maps the backbone's features and a normalized phase index directly to a whole action chunk in one forward pass, using roughly one-tenth—and for the largest backbone one-hundredth—of the parameters of the native head. We add an optional training-only **shape regularizer** (a topology-inspired pairwise-distance prior; not full persistent homology, §3.2). We drop the *identical* head into four backbones spanning more than 30× in size—ACT (88 M), SmolVLA (450 M), π0.5 (3.3 B), and GR00T N1.6 (3 B)—and test on the four LIBERO suites plus the seven-axis LIBERO-Plus robustness benchmark. The head matches or nearly ties the native baseline in-distribution on every backbone (essentially tying even on the frozen π0.5) while planning in a single forward pass (up to 5× lower latency on SmolVLA). Its robustness effect, however, is *conditional*, and this is our central observation: across the four backbones the operator head's robustness advantage tracks how much the backbone is allowed to adapt—from +20.6 points on adapting SmolVLA (LIBERO-Spatial, 5 seeds) and +3.3 on end-to-end ACT (suite-average, within noise) to −11.6 on partly-frozen GR00T and −13.4 on fully-frozen π0.5 (the π0.5 comparison trains both heads with identical photometric augmentation—the standard robustness lever for a frozen backbone—which halves but does not flip the gap). We report this as a consistent *trend*, not a law: with only four backbones the adaptation axis co-varies with model scale and training budget, and the decisive within-backbone experiment is left to future work (§8). **All evaluation is in simulation (LIBERO);** only the SmolVLA study is multi-seed, and we report the larger-backbone comparisons as single-run observations with 95% confidence intervals. We report the negative results in full, with audited provenance including a corrected closed-loop replan bug, and release all conditions and code.

Keywords: vision–language–action models, operator learning, DeepONet, action decoding, robustness, LIBERO, shape regularization, imitation learning.

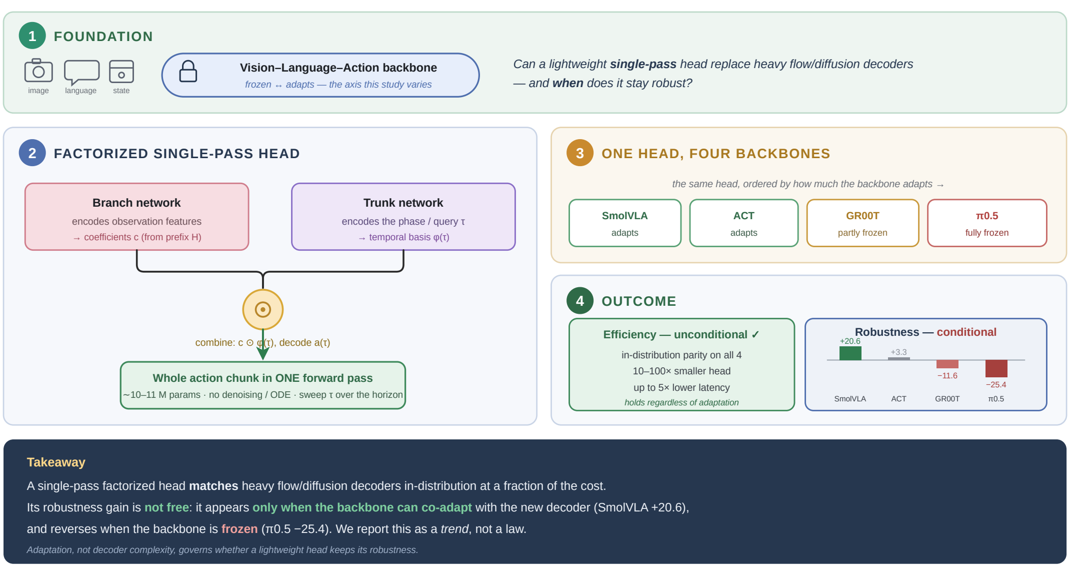

Figure 1. Graphical abstract. We replace a VLA model's heavy iterative flow/diffusion action head with a single-pass factorized (branch–trunk) head at ∼10–11 M parameters. In-distribution accuracy and efficiency improve unconditionally; the robustness effect is conditional and tracks how much the backbone is free to adapt (SmolVLA +20.6 → frozen π0.5 −13.4, the latter with photometric augmentation on both heads).

1. Introduction

Vision–language–action (VLA) models have become the dominant recipe for generalist robot manipulation. The idea is simple: a large pretrained vision–language model (VLM) acts as the *backbone*, reading multi-view camera images and a natural-language instruction and turning them into a rich, grounded representation. An *action head* then decodes this representation into a short horizon of continuous joint or end-effector commands, which the robot executes before looking again and re-querying the model.

The backbone holds the overwhelming majority of the parameters —and most of the semantic, spatial, and linguistic knowledge. Models such as OpenVLA [1], π0 [3], and GR00T [5] reuse large pretrained VLMs—from Prismatic/Llama-2 in OpenVLA to PaliGemma [34] in π0 and Eagle in GR00T—inheriting priors from billions of image–text pairs. The action head, by contrast, is where the policy actually commits to motion: it must turn high-dimensional context into precise, temporally coherent trajectories.

Almost every recent VLA uses a *generative* decoder for this last step. Diffusion policies [8], building on denoising diffusion [36], refine a noise sample over K denoising steps (typically 10–50). Flow matching [9], used in π0 and π0.5, replaces that discrete chain with a continuous ODE integrated by several Euler steps. Autoregressive chunking [7] predicts chunks with a conditional variational autoencoder (CVAE). All three trade compute for expressiveness: every control step needs several forward passes through the head, and the heads are large—π0.5's flow expert is about 300 M parameters, and GR00T N1.6's diffusion transformer (DiT) is 1.09 B. A parallel line of work reduces this cost with single-pass decoders—OpenVLA-OFT [19] regresses continuous actions in parallel, and FAST [20] compresses chunks with a learned tokenizer; our head shares that single-pass motivation but factorizes the decoder over a phase variable.

This paper asks whether a *single-pass, non-iterative* decoder can match these generative heads. We take an **operator-learning** view of action decoding and instantiate it with a DeepONet [11]-style branch–trunk network. The observation is that a demonstrator's action chunk can be seen as a *function* of a normalized phase variable $\tau \in [0,1]$ along the horizon, and the backbone's prefix tokens are the *input function* that decides which trajectory to produce. A branch–trunk factorization is a natural architectural prior for such a mapping—a branch network reads the input and a trunk network reads the query coordinate. We use this as motivation for the architecture, not as a theoretical guarantee (§3.1).

An operator head is small (about 10–11 M parameters, versus 100 M–1 B for generative heads) and fast (one forward pass, not 10–50 steps). But does it *generalize* under distribution shift? To probe this, we add an optional training-only shape regularizer (a topology-inspired pairwise-distance prior; §3.2), and we measure robustness directly with LIBERO-Plus [13], which systematically varies camera viewpoint, lighting, sensor noise, background texture, object layout, robot initial state, and language wording.

The heart of the paper is that we run the *same* head across four backbones—ACT (88 M), SmolVLA (450 M), π0.5 (3.3 B), and GR00T N1.6 (3 B)—that differ by more than 30× in size and, more importantly, differ in *how much of the backbone is allowed to change* during training. This turns a single architecture into a comparative study of *when* a factorized head helps—and the answer is that **adaptation, not decoder complexity, governs robustness transfer**: the head's robustness advantage grows with how much the backbone representation co-adapts with it (a consistent **co-adaptation trend**). We are deliberate about calling this a trend, not a law—our adaptation axis co-varies with scale and training budget, and the decisive within-backbone experiment remains future work (§8). We present this as a focused *empirical study*: the goal is not a new state-of-the-art system but a clear, honestly-scoped account of *when* a lightweight operator head helps or hurts, and why. Our contributions are:

1. **The main finding.** Through the first cross-backbone study of a single lightweight action head across four VLA families (88 M → 3.3 B), we identify **backbone adaptation—not decoder complexity—as the dominant factor** governing whether the head keeps its robustness under distribution shift: its advantage rises and falls with how much the backbone is free to co-adapt with it. We present this as a consistent empirical *trend*, not a causal law, and enumerate its confounds (scale, training budget) and the decisive within-backbone test that would settle it (§8).
2. **An unconditional efficiency win.** A drop-in **factorized branch–trunk head** (∼10–11 M params; inspired by operator learning but claimed only as an architectural prior, §3.1) replaces flow/diffusion decoders under an identical tensor contract and decodes a whole action chunk in a *single forward pass*. It matches the native head in-distribution on *every* backbone while cutting head size by up to ∼100× and planning latency by up to 5× (§5.5)—a benefit that holds regardless of adaptation.
3. **A practical decision rule, grounded in negatives.** Use the factorized head when the backbone adapts; keep the generative head when it is frozen. This rule rests on **fully documented negative results** on frozen-backbone π0.5 and partly-frozen GR00T, with audited provenance (including a corrected closed-loop replan bug) and open release of every training and evaluation condition.
4. A secondary training-only **shape regularizer** (a pairwise-distance spread prior, zero inference cost) whose effect we find *mixed*—helpful on long-horizon suites, mildly harmful on the SmolVLA headline—reported honestly rather than as a headline gain (§3.2).

2. Related Work

2.1. Vision–Language–Action Models

The VLA paradigm rests on the idea that large VLMs carry spatial and semantic priors that transfer to manipulation once we attach action prediction. RT-2 [32] first showed this by fine-tuning a PaLM-E [33] model to emit discretized actions as text tokens. OpenVLA [1] extended it to an open 7 B model with autoregressive discrete actions, and Octo [2] introduced a flexible multi-embodiment transformer. The π0 family [3,4] moved from discretization to continuous flow matching [9] on a PaliGemma [34] backbone; π0.5 added FAST tokenization [20] and open-world scale. NVIDIA's GR00T N1/N1.5/N1.6 [5] used a dual-system design inspired by fast/slow thinking: a mostly-frozen Eagle VLM (System 2) conditions a diffusion transformer (System 1). SmolVLA [6] targets affordability with a compact 450 M model. Most of these use generative heads; we instead swap only the decoder and hold the backbone interface fixed, which lets us study how decoder choice interacts with backbone adaptation (§4)—so our study necessarily *varies* how much the backbone changes rather than leaving it untouched.

2.2. Action Decoding Strategies

Action Chunking with Transformers (ACT) [7] uses a CVAE [35] to predict chunks, sampling a latent style variable for multimodality; at inference the latent is fixed, so ACT is itself effectively single-pass. Diffusion Policy [8] treats action generation as denoising over the chunk, needing K steps per query. Flow matching [9] replaces the discrete chain with a continuous ODE, usually integrated in about ten Euler steps. Both trade latency for expressiveness. Closest to us in spirit are single-pass decoders that drop iterative sampling entirely: OpenVLA-OFT [19] uses parallel continuous-action regression, and FAST [20] a learned action tokenizer. Relative to these, our head adds an explicit branch–trunk factorization over a phase variable and asks how such a decoder interacts with backbone adaptation and robustness.

Positioning among single-pass decoders. A conceptual comparison; a direct empirical benchmark against OpenVLA-OFT and FAST on LIBERO is future work, as their decoders are not drop-in under our tensor contract.

Property	OpenVLA-OFT [19]	FAST [20]	Ours
Decoding	parallel regression	tokenizer + AR	branch–trunk factorized
Explicit phase τ conditioning	no	no	yes
Cross-attention pooling	no	no	yes (m=8)
Single forward pass	yes	yes	yes
Cross-backbone robustness study	—	—	yes (LIBERO-Plus, 4 backbones)

2.3. Operator Learning

DeepONet [11], built on a universal approximation theorem for operators, learns maps between infinite-dimensional function spaces through a branch–trunk factorization. Fourier Neural Operators [12] do the same in the spectral domain, and physics-informed DeepONets [22] extend the idea to solution operators of parametric dynamical systems. These methods are mostly used to solve differential equations. We borrow the branch–trunk structure as an architectural prior for a different job—mapping a VLA backbone's context embedding (the "input function") to a continuous action trajectory (the "output function") indexed by phase τ—and we are explicit (§3.1) that our merge-and-decode step departs from the classical bilinear form, so the approximation theorem of [11] does not directly transfer.

2.4. Topology and Shape Priors in Learning

Persistent homology [14] gives a multi-scale topological summary of a point cloud—connected components (0-dim), loops (1-dim), and voids (2-dim) across scales. Differentiable topological losses, such as topological autoencoders [21], have been used to regularize representation models. Our shape regularizer (§3.2) is *inspired* by this line but is deliberately much simpler—a top-*k* pairwise-distance term—and we are careful (§3.2) not to claim it computes persistent homology; it captures coarse geometric spread rather than true 0-dimensional persistence. We are not aware of prior use of such a distance-spectrum shape prior directly on robot action chunks, but we treat this as a minor, secondary component of the paper.

2.5. Robustness Evaluation in Manipulation

LIBERO [10] provides four standardized manipulation suites of rising difficulty. LIBERO-Plus [13] extends them with seven systematic perturbation families for out-of-distribution testing. We treat LIBERO-Plus as our primary metric, because in-distribution success alone is a poor predictor of deployment reliability.

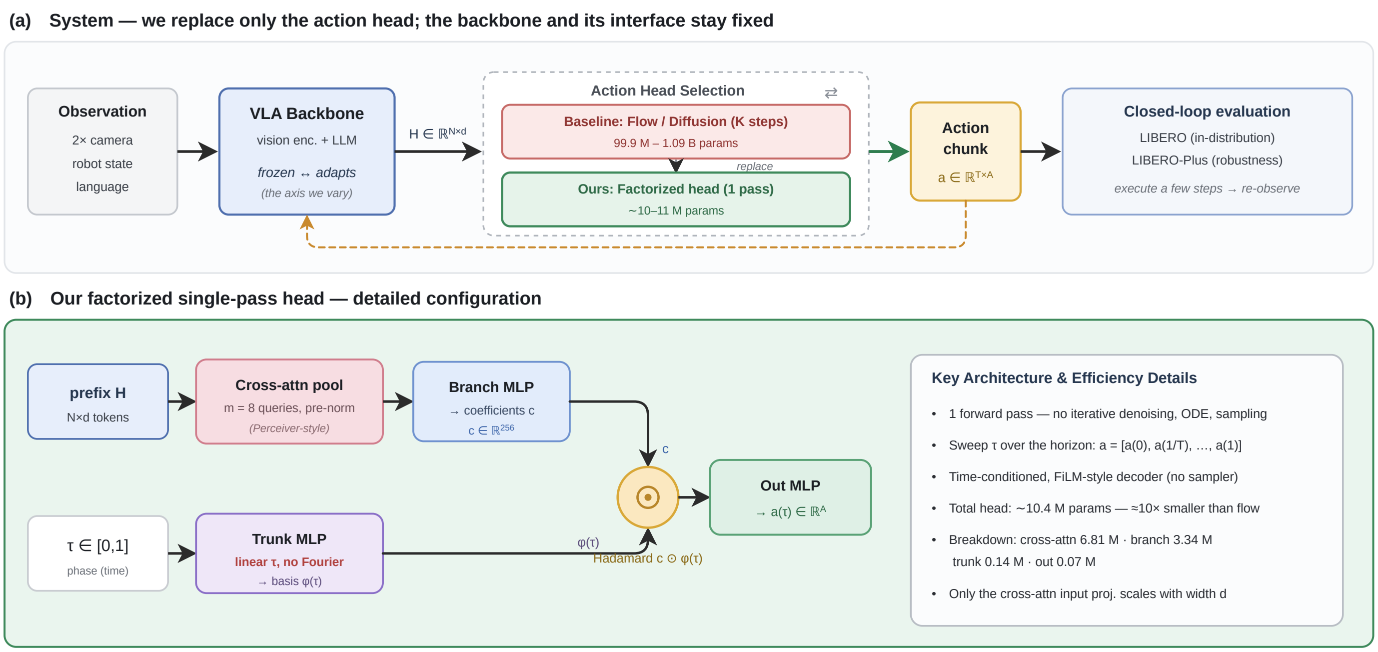

Figure 2. Architecture. (a) We swap only the action head; the backbone and its tensor interface are unchanged, so the identical study runs across every backbone. (b) The factorized single-pass head (the configuration fixed by the ablations of §6): a Perceiver-style cross-attention pool (m=8) and a branch MLP read the prefix tokens H into coefficients c; a linear trunk lifts the phase τ to a basis φ(τ); their Hadamard product is decoded to a(τ), and sweeping τ over the horizon yields the whole action chunk in a single forward pass. The optional training-only shape regularizer is described in §3.2.

3. Method

We keep the VLA backbone and its interface entirely fixed and replace only the action head. This is deliberate: it isolates the effect of the decoder and lets us study how head design interacts with backbone adaptation across models.

Problem setup. The backbone produces a set of prefix tokens $\mathbf{H}\in\mathbb{R}^{N\times d}$ that fuse vision, language, and (where available) proprioceptive state. The head must emit an action chunk $\mathbf{a}\in\mathbb{R}^{T\times A}$, where T is the horizon (number of future steps) and A is the action dimensionality (joint angles plus gripper). Baseline heads produce **a** with an iterative flow/diffusion sampler that needs K passes through the head; we produce it with a single evaluation. Fig. 2 shows the full architecture — the system-level head substitution (a) and the factorized head in detail (b) — and Fig. 3 gives a study overview.

3.1. The Operator-Style Action Head

Classical DeepONet [11] approximates a nonlinear operator $G:\mathcal{C}\to\mathcal{A}$ between function spaces as a finite-rank inner product of two networks—a branch that reads the input function and a trunk that reads the query coordinate. We adapt this *structure* to action decoding by treating the prefix tokens as the input function and the normalized phase $\tau=t/T\in[0,1]$ as the query. We stress that we use branch–trunk as an architectural prior: our merge step (a Hadamard product followed by a nonlinear decoder, Eq. 3) is not the classical bilinear sum, so the universal-operator-approximation guarantee of [11] does not carry over, and we make no theoretical claim from it—we evaluate the design empirically. The head has four stages.

Stage 1: Cross-attention pooling. The prefix tokens $\mathbf{H}\in\mathbb{R}^{N\times d}$ have a variable length N (it depends on image resolution, instruction length, and backbone), so we first compress them into a fixed-size summary. We use a Perceiver-style [16] cross-attention pool with m=8 learned query vectors $\mathbf{Q}\in\mathbb{R}^{m\times d_q}$ (d_q=512) that attend to $\mathbf{H}$ through 3 pre-norm [24] multi-head attention [23] blocks (8 heads each):

$$\mathbf{u} = \mathrm{CrossAttnPool}(\mathbf{Q}, \mathbf{H}) \in \mathbb{R}^{m\times d_q}. \tag{1}$$

The output is flattened to a single vector $\mathbf{u}$. This is the largest part of the head, 6.81 M parameters, and the only part whose input projection scales with the backbone width d. Cross-attention matters because it lets the head focus on the most action-relevant parts of the representation—object locations, gripper state, goal regions.

Stage 2: Branch network. A two-layer GELU [25] MLP maps the pooled summary to a coefficient vector:

$$\mathbf{c} = \mathrm{MLP}_{\mathrm{branch}}(\mathbf{u}) \in \mathbb{R}^{p}, \quad p = 256. \tag{2}$$

$\mathbf{c}$ encodes *what* trajectory to produce, distilled from the whole scene, instruction, and state. The branch has 3.34 M parameters.

Stage 3: Trunk network. The trunk maps the scalar phase τ to a basis vector $\varphi(\tau)\in\mathbb{R}^{p}$ with a two-layer GELU MLP (0.14 M parameters). It defines *when* in the horizon we are querying—a learned temporal basis that the branch's coefficients modulate. (An earlier version lifted τ into Fourier features [15]; our ablation in §6 shows a plain linear embedding is simpler and just as robust, so the final head drops the Fourier step.) At inference τ is swept over the fixed discrete horizon; we do not claim temporal super-resolution or resolution generalization.

Stage 4: Merge and decode. The branch coefficients and trunk basis are combined with a parameter-free Hadamard (element-wise) product, then decoded:

$$\mathbf{a}(\tau) = \mathrm{MLP}_{\mathrm{out}}(\mathbf{c} \odot \varphi(\tau)) \in \mathbb{R}^{A}, \tag{3}$$

where $\mathrm{MLP}_{\mathrm{out}}$ is a small two-layer network (0.07 M parameters). The full chunk is obtained by sweeping τ over the discrete horizon, $\mathbf{a} = [\mathbf{a}(0), \mathbf{a}(1/T), \ldots, \mathbf{a}((T-1)/T)] \in \mathbb{R}^{T\times A}$. Because the branch is evaluated once and the trunk/decode are trivially cheap, the entire chunk is produced in a **single forward pass**—no denoising loop, no ODE, no sampling. Functionally this is a time-conditioned (FiLM-style) decoder with a branch–trunk prior; we adopt the operator framing for intuition, not as a claim beyond what the ablations support. The total count is ∼10.4 M for SmolVLA (d=960) and ∼11 M for π0.5/GR00T (d=2048).

3.2. A Training-Only Shape Regularizer

Mean-squared error makes the head reproduce individual action values, but says nothing about the *shape* of the trajectory. Two chunks can have similar MSE yet differ in smoothness, spacing, or spread. We add an optional training-only regularizer that penalizes coarse geometric differences between the predicted and expert chunks.

Intuition. We treat a chunk $\hat{\mathbf{a}}=\{\hat{\mathbf{a}}_1,\ldots,\hat{\mathbf{a}}_T\}\subset\mathbb{R}^{A}$ as a cloud of T points in action space and match a summary of its overall geometric extent to the demonstrator's. This nudges the head toward the right coarse geometry, not just the right pointwise values.

Differentiable surrogate. We use a light, fully differentiable summary: the sorted top-k values of the pairwise-distance matrix. With $\mathrm{pdist}(\mathbf{x})$ the vector of all pairwise Euclidean distances among the T points,

$$\mathcal{L}_{\mathrm{SR}} = \| \mathrm{top}_k(\mathrm{pdist}(\hat{\mathbf{a}})) - \mathrm{top}_k(\mathrm{pdist}(\mathbf{a}^*)) \|_1, \tag{4}$$

where top_k keeps the k=8 *largest* distances and $\mathbf{a}^*$ is the expert chunk. The total loss is $\mathcal{L} = \mathcal{L}_{\mathrm{MSE}} + \lambda\,\mathcal{L}_{\mathrm{SR}}$ with λ=0.02. **A note on terminology.** This term is *inspired by* topological data analysis but is *not* persistent homology: 0-dimensional Vietoris–Rips persistence is given by the minimum-spanning-tree edge lengths (the *smallest* merge distances), whereas Eq. 4 matches the *largest* pairwise distances, i.e. the trajectory's diameter/spread. We use the largest distances because they are cheap, stable to optimize, and empirically the most useful coarse shape cue; we do not claim a topological invariant. The term adds **zero parameters** and **zero inference cost** (Fig. 2c); it is a pure loss computation applied only during training. As we show (§6–7), its effect is mixed, so we treat it as a secondary component.

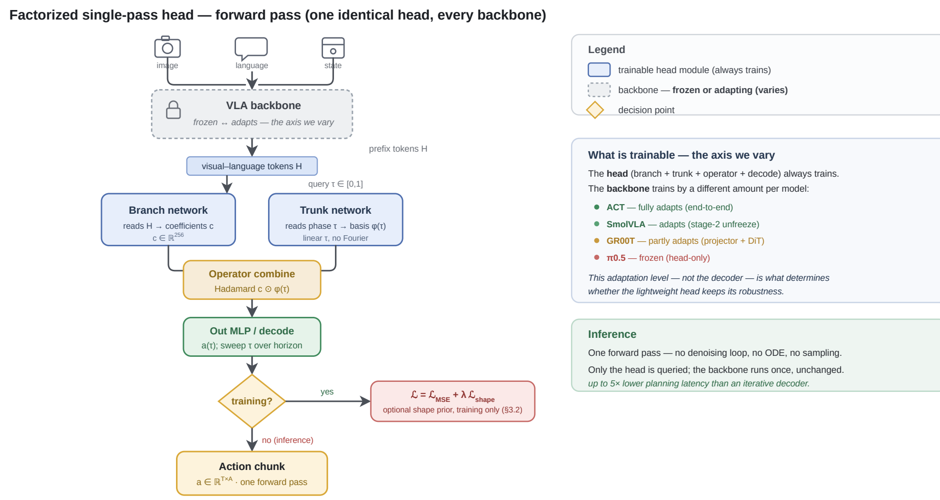

Figure 3. Forward pass of the factorized head. The VLA backbone encodes the inputs into prefix tokens H; a branch network maps H to coefficients c and a trunk network maps the phase τ to a basis φ(τ); their Hadamard product is decoded, and sweeping τ over the horizon yields the whole action chunk in a single pass. The optional shape prior is training-only (§3.2). The head always trains, while the backbone is frozen or adapting depending on the model—and this adaptation level, not the decoder, is what governs robustness transfer.

3.3. Backbone Integration

The head is a true drop-in: it reads the same prefix-token tensor **H** and returns the same chunk tensor **a** as the native head, with no change to the backbone, data pipeline, or evaluation harness. For SmolVLA and π0.5 we set T=50, A=32; for GR00T, T=16, A=29, with the robot state added as an extra prefix token. One code path implements all four integrations; only the input-projection width changes (d=960 for SmolVLA, d=2048 for π0.5/GR00T). This uniformity is what lets us hold the decoder fixed while varying the backbone by 30×—the core of our study.

Integrating the head into a VLA (drop-in)

(1) Expose the backbone's prefix tokens **H** (width *d*). (2) Cross-attention pool: *m*=8 queries, 3 pre-norm blocks. (3) Branch MLP: pooled summary → coefficients **c** (p=256). (4) Trunk MLP: phase τ → basis φ(τ). (5) Decode $\mathbf{a}(\tau)=\mathrm{MLP}_{\mathrm{out}}(\mathbf{c} \odot \varphi(\tau))$ and sweep τ over the horizon for the whole chunk. **Training:** keep the backbone's native optimizer and loss (MSE, plus the optional shape prior). **Only** the cross-attention input projection changes with *d*—every other module is backbone-agnostic.

4. Experimental Setup

Every number in this paper — including all flow, diffusion, and CVAE baselines — is our own reproduction, run under identical conditions with closed-loop simulated evaluation; we quote no result from prior work. Every operator-vs-baseline comparison is *symmetric within a campaign*: identical data, schedule, and hyperparameters, with the heads differing only in architecture and the shape-regularizer (SR) flag, so any within-campaign difference is attributable to the decoder alone. The SmolVLA study uses 5 seeds with significance tests; the larger-backbone campaigns are single-run, so we read the cross-backbone comparison as a trend rather than a head-to-head (§7–8). **Implementation.** All models are implemented in PyTorch [30] and trained with AdamW [26,27], a cosine learning-rate schedule [28], and bf16 mixed-precision [29]; the simulator is MuJoCo [31], accessed through the LIBERO/robosuite [37] stack. Each campaign runs on a single GPU; across the project we used NVIDIA Blackwell, A100, and A6000 cards.

Table 1. Backbones and the identical factorized (branch–trunk) head. Its parameter count scales only through the cross-attention pool's input projection.

Model	Total	Vision	Native head	Ours
ACT	88 M	ResNet-18	Transformer dec. 37.8 M	10.2 M
SmolVLA	∼450 M	SigLIP [17]	Flow expert 99.9 M	10.4 M
π0.5	∼3.3 B	SigLIP-So400m	Flow expert ∼300 M	∼11 M
GR00T N1.6	∼3 B	Eagle	DiT 1.09 B	∼11 M

4.1. Backbones

We pick four backbones spanning two orders of magnitude in size and—crucially—differing in how much of the backbone adapts (Table 1). This is the axis the study turns on, and we note upfront that it is *correlated* with model size and training budget, a confound we return to in §7–8. **ACT** (88 M) is trained end-to-end from scratch—the backbone fully adapts. **SmolVLA** (450 M) uses a two-stage recipe: stage-1 trains the head alone, stage-2 unfreezes the backbone at a low learning rate—partial adaptation. **π0.5** (3.3 B) is fine-tuned head-only with a frozen backbone—no adaptation. **GR00T N1.6** (3 B) tunes the projector and the DiT while keeping the VLM mostly frozen—partial adaptation of a different kind.

4.2. Benchmarks

We evaluate on the four LIBERO suites [10]—Spatial (spatial reasoning), Object (object recognition), Long (`libero_10`, long-horizon), and Goal (goal-conditioned)—measuring in-distribution success. For robustness we use LIBERO-Plus [13], which perturbs each task along seven independent axes: (1) camera viewpoint, (2) lighting, (3) sensor noise, (4) background texture, (5) object layout, (6) robot initial state, and (7) language paraphrasing. Robustness is the mean success over the seven categories.

4.3. Training Regimes and Evaluation Protocol

Table 2 lists the exact recipe per campaign. All evaluation is closed-loop and receding-horizon. In-distribution: 10 tasks × multiple episodes with varied initial states. LIBERO-Plus: 7 categories × 8–15 episodes, stratified by a fixed seed. Because episode counts are modest, per-category numbers carry wide binomial intervals and should be read as coarse; we therefore emphasize suite-averages and consistent patterns over individual cells. The within-model comparison is symmetric; cross-model absolute numbers carry the replan/horizon differences described below, so we read them as trends rather than head-to-head.

Table 2. Training conditions per campaign. Every DeepONet-vs-baseline comparison is symmetric within a campaign (identical data, schedule, hyperparameters); the heads differ only in architecture and the shape-regularizer (SR) flag. “BB” = backbone.

Campaign	Regime	BB	Steps	Batch
SmolVLA	stage-2 unfreeze	adapts	1.65K / 6.65K	48
ACT	end-to-end	adapts	30K	64
π0.5	frozen, head-only	frozen	15K→15K	16
GR00T	projector + DiT	part. frozen	15K→15K	16×4

5. Results

We report each backbone in turn. Throughout, **ind** = in-distribution success (%) and **Plus** = LIBERO-Plus robustness (%); best robustness per suite is **bold**. "SR" denotes the shape-regularizer variant of §3.2.

5.1. SmolVLA (450 M) — the origin study

On LIBERO-Spatial with 5 seeds, the operator head *ties* the flow baseline in-distribution (80.3 vs 79.4, p=0.61, not significant) but **more than doubles** robustness (38.5 vs 17.9), winning 6 of 7 perturbation categories at 5× lower latency (Table 3). Per category (Flow→DeepONet): Layout 9.3→48.0 (+38.7), Language 8.0→33.3 (+25.3), Lighting 22.7→61.3 (+38.6); the flow baseline wins only Sensor Noise. This is the one campaign with multi-seed significance, and the robustness effect is large relative to its variance. The gains concentrate on *geometric* perturbations (layout, viewpoint, initial state) rather than photometric ones, which we read as a hypothesis—the co-adapted representation may learn more viewpoint-invariant features when trained with the operator head—rather than a settled mechanism.

Table 3. SmolVLA, LIBERO-Spatial, 5 seeds (mean). Latency = planning time per chunk. Flow baseline reproduced by us.

Head	ind %	Plus %	Latency	Params
Flow (baseline)	79.4	17.9	148.1 ms	99.9 M
Ours (factorized)	80.3	**38.5**	29.5 ms	10.4 M
Ours + SR	81.5	32.6	29.5 ms	10.4 M

5.2. ACT (88 M) — end-to-end, all four suites

Trained end-to-end from scratch (30 K per suite), the operator head again helps on average (Table 4): it improves the whole-suite average on both axes (ind +4.0, robustness +3.3) and wins Spatial, Object, and Goal robustness. The direction is consistent with an adapting backbone helping the operator head; we read the +3.3 average delta as directional support rather than a large effect, comparable to the run-to-run robustness spread we measure on SmolVLA (±6.9, Table 8). Bootstrap 95% confidence intervals (10 k resamples over the 84 LIBERO-Plus instances per suite) put every per-suite ACT delta's interval across zero—Spatial +3.6 [−10.7, +17.9], Object +7.1 [−8.3, +22.6], Long −2.4 [−15.5, +10.7], Goal +4.8 [−10.7, +19.0]—so we treat ACT as a directional, not significant, data point.

Table 4. ACT (from-scratch, 30 K/suite). In-distribution / LIBERO-Plus success (%). ACT baseline reproduced by us.

Suite	ACT		Ours		+SR	
	ind	Plus	ind	Plus	ind	Plus
Spatial	83.7	61.9	83.7	**65.5**	78.0	57.1
Object	81.7	46.4	87.0	**53.6**	72.0	47.6
Long	54.7	22.6	62.7	20.2	50.0	**27.4**
Goal	83.3	54.8	86.3	**59.5**	85.0	51.2
Avg	**75.9**	**46.4**	**79.9**	**49.7**	**71.3**	**45.8**

5.3. π0.5 (3.3 B) — frozen backbone: a negative result

Scaling to π0.5 with a frozen backbone and head-only fine-tuning, the operator head **loses**: with identical photometric augmentation on both heads (our final π0.5 protocol; see the provenance note), the flow baseline ties on the in-distribution *average* (+0.2) but leads on robustness (+13.4 on average). The signature is diagnostic (Table 5): in-distribution the readout has enough capacity to fit the training data regardless of head—the operator head even *wins* in-distribution on three suites (Spatial/Object/Goal, 92–98%)—but LIBERO-Plus robustness trails. We flag one caveat behind the in-distribution “tie”: the average hides a real per-suite gap on Long (operator 53.3 vs flow 70.0), where the operator head is clearly under-trained under the head-only budget. The shape regularizer does *not* rescue this: **+SR is worse still** (Plus 36.6, a −21.0 gap; in-distribution 78.5), the opposite of its helpful effect on the adapting backbones (ACT/SmolVLA)—consistent with the reading that a training-only shape prior cannot reshape a frozen representation. Augmentation, the standard way to buy robustness on a frozen backbone, lifts both heads and roughly halves the gap versus the un-augmented run (−25.4→−13.4) but does not flip its sign.

Interpretation (a hypothesis). The π0.5 backbone was pretrained end-to-end as a flow model, so its internal features are tuned for the flow head's multi-step denoising. Bolting on an operator head *without letting the backbone adapt* may leave those features “foreign” to the new decoder: the head can memorize the training distribution (near-parity in-distribution) but reuses less of the backbone's prior for out-of-distribution generalization. We offer this as an interpretation, not a proven mechanism—note that this comparison is *flow-vs-operator on an identically frozen backbone*, so it evidences a head-generalization gap under freezing; converting it into a controlled adaptation manipulation is the decisive test we flag in §8.

Provenance note. The π0.5 campaign was run under three successively-corrected protocols, and we report the last as canonical. (i) An original run used the config default `n_action_steps=50` (effectively open-loop) instead of `replan=5` (genuine closed-loop); (ii) after fixing the replan bug (and an asymmetric `--init`) we re-ran at replan=5 (flow 54.9 vs Ours 29.5 on Plus, a −25.4 gap); (iii) we then re-ran once more with identical per-sample photometric augmentation on all heads—the standard robustness lever for a frozen backbone—which is the canonical result in Table 5. The ordering is *invariant* across all three protocols: flow leads robustness on the frozen backbone (27.7 vs 16.1 open-loop; 54.9 vs 29.5 replan=5; **57.6 vs 44.2 with augmentation**). The negative result is a property of the frozen head-only regime, not of any single protocol.

Table 5. π0.5, canonical closed-loop replan=5 **with photometric augmentation** (`--color_jitter`, identical for all heads). ind / Plus success (%), single seed. Flow baseline reproduced by us.

Suite	Flow		Ours		+SR	
	ind	Plus	ind	Plus	ind	Plus
Spatial	87.5	**71.4**	92.5	44.6	90.8	35.7
Object	88.3	**57.1**	92.5	55.4	85.8	53.6
Long	70.0	**46.4**	53.3	26.8	44.2	16.1
Goal	90.8	**55.4**	97.5	50.0	93.3	41.1
Avg	**84.2**	**57.6**	**84.0**	**44.2**	**78.5**	**36.6**

5.4. GR00T N1.6 (3 B) — a partly-frozen middle case

GR00T completes the picture, and it is the most interesting case because its backbone is *partly* frozen: the projector and the diffusion transformer train, while the Eagle VLM is mostly frozen. The diffusion baseline is very strong (Table 6): about 98% in-distribution and up to 91% robustness. This is expected of a 3 B pretrained backbone that is inherently appearance-robust: the diffusion Spatial breakdown shows Camera 62.5 and Robot-init 75.0, while Light, Sensor, Texture, Layout, and Language all score 100.0—an *appearance-robust, geometry-fragile* signature. We verified these high scores are not a skipped-perturbation artifact by cross-checking against ACT, which drops to ~58% on the same photometric axes; we note the small per-category episode counts still make the exact values coarse.

Head-to-head. In-distribution, the operator head essentially *ties* the 1.09 B DiT (96.9 vs 97.9 on average) while using about 11 M parameters—a roughly hundred-fold smaller head. On robustness, however, the operator head loses (75.9 vs 87.5), and +SR partly recovers it (78.1), mainly by lifting Object-Plus from 75.0 to 82.1. This is the same qualitative signature we saw on π0.5—in-distribution parity, robustness gap—but the gap is *smaller* (−11.6 vs −13.4), and GR00T lets more of its representation adapt. GR00T therefore sits as the middle point of the trend in Fig. 4, with the caveat that it also differs from π0.5 in size and horizon.

Table 6. GR00T N1.6-3B, closed-loop horizon=8. ind / Plus success (%). Diffusion baseline reproduced by us. Campaign complete (all 3 variants × 4 suites).

Suite	Diffusion		Ours		+SR	
	ind	Plus	ind	Plus	ind	Plus
Spatial	98.3	**91.1**	96.7	83.9	98.3	83.9
Object	98.3	**83.9**	99.2	75.0	99.2	82.1
Long	95.8	**91.1**	95.0	69.6	87.5	73.2
Goal	99.2	**83.9**	96.7	75.0	97.5	73.2
Avg	**97.9**	**87.5**	**96.9**	**75.9**	**95.6**	**78.1**

5.5. Efficiency

The operator head is decisively cheaper where the baseline is iterative (Table 7). The saving comes from removing the multi-step sampling loop: one forward pass instead of 10–50. On SmolVLA the head plans an action chunk 5× faster (148.1 → 29.5 ms plan latency). On ACT the gain is small (16.0 → 13.4 ms, 1.2×) precisely because ACT's CVAE is already single-pass—confirming the speed-up comes from removing iterative sampling, not from the operator design per se. On π0.5 we report a *training*-step saving (0.818 → 0.385 s per optimizer step, 2.1×); we did not separately measure π0.5/GR00T inference latency and do not claim a specific inference speed-up there. On parameters, the head is ~10× smaller than SmolVLA's 99.9 M flow expert and ~100× smaller than GR00T's 1.09 B DiT—the latter a ~36% reduction in *total* model size—at in-distribution parity. Uniformly across the four backbones, the ~10–11 M head replaces native decoders of 37.8 M (ACT), 99.9 M (SmolVLA), ∼300 M (π0.5), and 1.09 B (GR00T)—head-size reductions of 3.7×, 9.6×, ∼27×, and ∼99× respectively. We keep the latency and parameter claims separate rather than merging them into a single headline multiplier.

Table 7. Efficiency, partitioned into inference and training. Head params = decoder parameters; reduction is head-only. Plan latency = planning time per chunk (batch 1, bf16); train step = median wall-time per optimizer step. π0.5 and GR00T run wrapped production inference pipelines from which we could not cleanly isolate decoder latency, so we report their training-step saving instead; metrics are never merged across rows.

Backbone · head	Head params	Reduction	Plan latency	Train step
SmolVLA · flow	99.9 M	—	148.1 ms	—
SmolVLA · ours	10.4 M	9.6×	29.5 ms (5.0×)	—
ACT · CVAE	37.8 M	—	16.0 ms	—
ACT · ours	10.2 M	3.7×	13.4 ms (1.2×)	—
π0.5 · flow	∼300 M	—	not meas.	0.818 s
π0.5 · ours	∼11 M	∼27×	not meas.	0.385 s (2.1×)
GR00T · DiT	1.09 B	—	not meas.	—
GR00T · ours	∼11 M	∼99×	not meas.	—

Key findings

- **Efficiency and in-distribution parity are unconditional.** The factorized head matches native flow / diffusion / CVAE decoders in-distribution on all four backbones, at up to ∼100× fewer head parameters and up to 5× lower planning latency.
- **Robustness is conditional.** It improves where the backbone adapts (SmolVLA +20.6; ACT +3.3, n.s.) and degrades where the backbone is frozen (GR00T −11.6; π0.5 −13.4, with augmentation).
- **The ordering variable is backbone adaptation**—not decoder architecture or head size. The lightweight head helps exactly when the representation is free to co-adapt with it.
- **Practical rule.** Use the factorized head when the backbone adapts; keep the generative head when it is frozen.

6. Ablation Study

We ablate the head on SmolVLA LIBERO-Spatial (Table 8, 3–5 seeds) to isolate each part. We report std alongside each configuration, and read the robustness column cautiously because its variance is large (±5–9 points).

Basis dimension p. Cutting p from 256 to 64 keeps in-distribution (81.7) but lowers robustness (32.7 vs 38.5), pointing to the richer basis capturing finer trajectory structure—though the intervals overlap.

Fourier features. Removing the Fourier lift (plain linear τ) barely changes robustness (39.0 vs 38.5) but raises variance (±8.7 vs ±6.9). Fourier stabilizes training without driving the result—so our *final* deployed head (Fig. 2) uses the simpler linear embedding. For provenance: the headline SmolVLA number in Table 3 (38.5) is the Fourier configuration of this ablation, and the deployed linear head scores 39.0 here—a within-noise difference that does not affect any conclusion.

Cross-attention depth. Dropping from 3 blocks to 1 hurts both axes (ind −2.1, Plus −7.7): the multi-block pool is important for compressing variable-length prefix tokens.

Regression baseline. Replacing the operator factorization with a plain regression MLP (same pooling, same budget) recovers in-distribution (83.0) but is lower on robustness (32.7 vs 38.5). This is the ablation that asks whether the branch–trunk *structure*, not merely the small parameter count, drives robustness. We read it as *suggestive*: the +5.8-point gap is within the full head's own std (±6.9), and holds on a single suite and backbone, so we do not claim it is conclusive—a multi-suite, multi-seed version is needed to settle it.

Table 8. Ablation on SmolVLA LIBERO-Spatial. Mean ± std over 3–5 seeds.

Configuration	ind %	Plus %
Full factorized head (p=256, 3 blk, Fourier)	80.3 ± 2.4	**38.5 ± 6.9**
(−) basis p: 256 → 64	81.7 ± 0.5	32.7 ± 5.2
(−) Fourier-τ (linear only)	82.7 ± 1.4	39.0 ± 8.7
(−) cross-attn blocks: 3 → 1	78.2 ± 2.0	30.8 ± 4.5
Regression head (no operator)	83.0 ± 4.2	32.7 ± 0.9

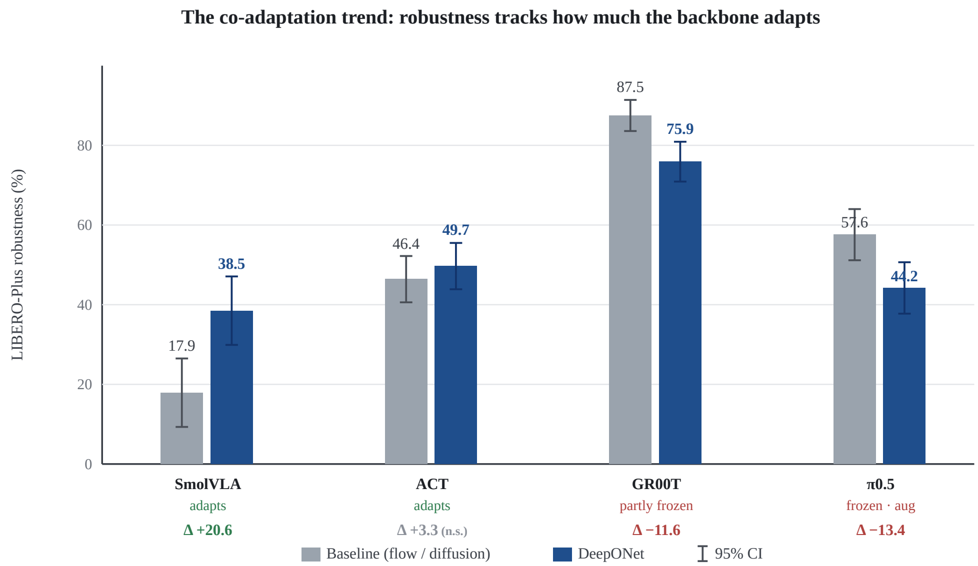

Figure 4. The co-adaptation trend, with 95% confidence intervals. LIBERO-Plus robustness of the baseline (grey) vs the operator head (blue) across four backbones ordered by adaptation. Error bars are 95% intervals: across-seed for SmolVLA (5 seeds); binomial (Wilson) at the evaluation episode count for the single-run backbones. The SmolVLA gain and the GR00T loss exceed their intervals; the π0.5 loss (Δ −13.4, both heads with photometric augmentation) is borderline—its 95% intervals nearly touch (baseline [51.0, 63.9] vs operator [37.8, 50.7], n=224)—and the ACT +3.3 does not reach significance (n.s.). SmolVLA is Spatial-only; ACT/π0.5/GR00T are suite-averages. Δ is operator minus baseline.

7. Discussion

The co-adaptation trend. We deliberately read Fig. 4 as a *sign split*, not a smooth ordering, because only two of the four deltas are individually significant. The head **gains** robustness where the backbone *adapts* (SmolVLA stage-2 unfreeze +20.6, the one multi-seed point, $p<0.001$; ACT end-to-end +3.3, *not* significant) and **loses** robustness where the backbone is *frozen or mostly frozen* (GR00T −11.6, π0.5 −13.4). After augmentation the two frozen-cluster points sit within ∼1.8 pp of each other, so we do *not* claim a monotone GR00T-vs-π0.5 ordering—only that both fall on the "loses" side of the adapt/frozen split. We call this a *trend*, not a law, and are explicit about the confounds: with four backbones the adaptation axis co-varies with model size (88 M → 3.3 B), native-head family, and training budget, so an equally consistent reading is "the small-head robustness loss appears where the baseline is already strong"; and one further inconsistency is ours to own—the π0.5 point uses photometric augmentation on both heads while the other three do not, so Fig. 4 mixes one augmented backbone with three un-augmented ones (each backbone's within-recipe Δ is still a fair flow-vs-operator measurement, and augmenting π0.5 only *shrinks* its loss, but the trend axis is not a single controlled intervention). The proposed mechanism—robustness lives in the representation, in-distribution capacity in the readout—is a hypothesis we have not tested with representation-level probes. What would move this from trend to law is a single-backbone experiment that varies *only* adaptation (§8).

The role of the shape regularizer. SR is a mixed signal. It improves in-distribution accuracy on SmolVLA (+1.2) but lowers robustness there (−5.9 vs vanilla DeepONet). Yet on the hardest long-horizon settings it is the best variant—ACT Long (27.4 vs 20.2) and GR00T Object-Plus (82.1 vs 75.0). We read this cautiously: the coarse spread prior may help most when the action space is complex and multi-modal (long-horizon, many objects) but over-regularize an already well-conditioned policy. Given the mixed picture, we present SR as a secondary component, not a headline contribution.

8. Limitations and Future Work

- **The decisive co-adaptation test (the key one).** The trend is confounded with scale and budget. Unfreezing the *same* π0.5 (or SmolVLA) backbone at 2–3 freeze levels on 1–2 suites—holding family, size, and data fixed—would vary only adaptation and turn the trend into a controlled result. We leave this experiment to future work; until it is done we deliberately frame the finding as a trend.

- **Statistical breadth.** Only SmolVLA-Spatial is multi-seed with a significance test; the other campaigns are single-seed, and several key deltas are within the run-to-run std we measure (±5–9 robustness points). LIBERO-Plus uses 8–15 episodes per category, so per-category values are coarse. Multi-seed runs and larger episode counts are the natural next step; to be transparent about the current basis we attach 95% confidence intervals to the headline figure (Fig. 4)—though for the single-run backbones these reflect only within-run episode sampling, not training-seed variance, and so understate the true uncertainty. Readers should weigh suite-averages and consistent patterns over individual cells.
- **Method scope.** We use the operator/DeepONet framing as an architectural motivation, not a theoretical result, and the shape regularizer is a coarse spread prior, not persistent homology (§3.2). We do not demonstrate operator-specific properties (e.g., temporal super-resolution).
- **Evaluation scope.** All evaluation is simulated (LIBERO); no physical-robot results are included. Cross-model absolute comparisons carry a replan/horizon difference (5 vs 8); within-model comparisons are symmetric.

9. Conclusion

We replaced the generative action decoder of four VLA models with a single-pass, operator-style DeepONet head and an optional training-only shape regularizer. The head is about 10× smaller than typical generative heads—and up to 100× smaller than GR00T's diffusion transformer—and up to 5× faster to plan where the baseline is iterative, yet it matches or nearly ties the native baseline in-distribution on every backbone we tested (within ~2.5 points at worst, on the frozen π0.5). Its robustness benefit is conditional: across our four backbones it tracks how much the backbone adapts, from +20.6 points on adapting SmolVLA to −13.4 on frozen π0.5 (with augmentation), with partly-frozen GR00T's −11.6 in between. We report this as a consistent trend rather than a law, because our four-backbone axis is confounded with scale and budget; the single-backbone unfreeze experiment that would make it causal, together with multi-seed statistics, remains future work. We report the negative results in full, with audited provenance including a corrected closed-loop replan bug, and release every training and evaluation condition. Even where robustness does not improve, the head's unconditional wins—in-distribution parity at a fraction of the parameters and a single-pass decode—make it a practical, honestly-scoped tool. Our broader takeaway is that *decoder efficiency alone does not determine robustness*: across four backbones it is the adaptation of the underlying VLA—not the choice of decoder—that governs whether a lightweight action head can replace an iterative one, and we hope this reframing helps the community study head design and backbone optimization together rather than in isolation.

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